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WaitLess: A Digital Queue Management System for Zimbabwean Public Hospitals

Case Study: Chinhoyi Provincial Hospital

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Date: June 2026

A PROJECT SUBMITTED IN PARTIAL FULFILMENT
OF THE REQUIREMENTS FOR THE BACHELOR OF
SCIENCE HONOURS DEGREE IN INFORMATION TECHNOLOGY

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CHAPTER 1: INTRODUCTION

1.0 Introduction

Healthcare delivery in public hospitals is significantly impacted by the efficiency of patient management systems. In Zimbabwe, many public health facilities continue to rely on manual, paper-based processes for patient registration, triage, and queue management. These outdated processes result in long waiting times, patient congestion, poor service delivery, and increased risk of adverse health outcomes. As Zimbabwe pursues its Vision 2030 agenda and seeks to modernise public service delivery, there is an urgent need to integrate digital technologies into the healthcare system.

This chapter presents the development of WaitLess, a digital queue management system designed specifically for Zimbabwean public hospitals. The system is informed by a physical case study conducted at Chinhoyi Provincial Hospital, where patient flow processes, triage protocols, departmental workflows, and patient origin patterns were documented and analysed. The case study revealed critical inefficiencies in the hospital's outpatient department (OPD) flow, casualty department triaging, and opportunistic infection clinic management.

A key feature of WaitLess is its integrated patient notification system, which will alert patients when their turn is approaching through two channels: WhatsApp notifications for patients who provide a contact number at registration, and real-time web push notifications for patients who registered through the web application. This addresses a critical gap identified during site observations: patients who leave the hospital premises while waiting have no way of being notified, leading to no-shows and queue disruption. The WhatsApp channel is designed to operate at no cost to the hospital by leveraging Meta's service conversation policy, whereby a patient who sends an initiating message to the system's WhatsApp Business number at registration opens a free 24-hour messaging window within which all subsequent queue notifications are delivered without charge.

This chapter is organised as follows: Section 1.1 presents the background to the study, outlining the broader context of queue management in healthcare and the specific challenges observed in the Zimbabwean setting. Section 1.2 articulates the problem statement, contrasting the ideal situation with the reality observed at Chinhoyi Provincial Hospital and the resulting consequences. Section 1.3 states the aim and objectives of the project. Section 1.4 discusses the significance of the study, while Section 1.5 outlines its limitations and delimitations. Section 1.6 defines key terms used throughout the chapter, and Section 1.7 concludes with a summary and a preview of the next chapter.

1.1 Background of the study

Queue management in healthcare is defined as the systematic process of controlling the flow of patients from entry to exit within a healthcare facility (Preater, 2002). Effective queue management is critical to the functioning of any hospital, as it directly influences patient satisfaction, clinical outcomes, and staff productivity. In high-income countries, digital queue management systems (DQMS) are widely deployed, incorporating features such as electronic patient registration, automated triage prioritisation, real-time wait-time displays, and digital notifications to patients (Cayirli & Veral, 2003).

In Sub-Saharan Africa, including Zimbabwe, the healthcare system faces unique challenges. According to the World Health Organization (2019), the African region bears over 24% of the global disease burden while having access to only 3% of health workers and less than 1% of world health expenditure. The resulting strain manifests in overcrowded outpatient departments, inefficient manual patient registration, and chaotic queuing that can endanger critically ill patients who do not receive timely care.

Zimbabwe's public hospitals have historically operated with paper-based patient records and manual queue management. Research in comparable Sub-Saharan African contexts has found that patients at public hospitals spend several hours waiting for outpatient consultations, with the majority of this time spent in unmanaged queues (Egbunu et al., 2020). This is consistent with findings from other low- and middle-income countries where manual queuing systems lead to inequitable service delivery, with vulnerable and critically ill patients sometimes being overlooked (Bidari et al., 2021).

The case study conducted at Chinhoyi Provincial Hospital corroborated these findings. The hospital operates a colour-coded triage system categorised into four priority levels: Priority I (Red) for critically ill patients such as those with respiratory arrest or severe burns, directed to the Casualty Resuscitation Room; Priority II (Yellow) for seriously ill but non-life-threatening cases directed to the Casualty Observation Ward; Priority III (Green) for ambulant patients with minor injuries directed to the Casualty Waiting Area; and Priority IV (Black) for deceased patients held until transfer to the mortuary. Despite this structured triage framework, the absence of a digital system to support these processes means that patient sorting relies entirely on manual observation and handwritten records.

The hospital's OPD Patient Flow Map revealed a structured but entirely paper-based pathway: patients began at registration (5 minutes), proceeded to nurse observation (5 minutes) and nurse consultation (10 minutes), waited at the doctor's queue line (10 minutes), then underwent doctor's consultation (10 minutes) before branching to laboratory (20 minutes), radiology (30 minutes), or direct admission. Results were then reviewed by a doctor (5 minutes), followed by pharmacy (10 minutes), the treatment room (10 minutes), and departure. The total expected journey time for a straightforward OPD visit was thus approximately 80 minutes under ideal conditions. In practice, the absence of digital tracking means bottlenecks are invisible to management and cannot be proactively addressed.

The hospital also operates an Opportunistic Infection (OI) Clinic, which serves HIV/AIDS patients requiring TB services, antiretroviral therapy (ART) resupply, viral load testing, cervical cancer screening, and counselling. Managing these multiple parallel service streams without a digital system creates confusion, missed appointments, and inequitable service distribution.

Digital queue management systems have been successfully implemented in hospitals across Sub-Saharan Africa. Mboya et al. (2024) reported that digital healthcare interventions, including electronic medical records and workflow optimisation strategies, significantly

reduced outpatient waiting times at Kilimanjaro Christian Medical Centre in Tanzania. In South Africa, the Chris Hani Baragwanath Hospital implemented a hybrid digital-physical queue system in its outpatient pharmacy, reporting significant improvements in patient satisfaction scores (Coetzee et al., 2020). These precedents demonstrate the feasibility and impact of digital queue management implementations in resource-constrained African hospital settings.

Despite the availability of such technologies, their adoption in Zimbabwe's public hospitals has been negligible. A key barrier is the lack of locally developed, context-appropriate solutions that account for infrastructural realities such as intermittent internet connectivity, limited patient digital literacy, and constrained hospital IT budgets (Gombe et al., 2019). This project seeks to address this gap by designing WaitLess specifically tailored to the operational context of Chinhoyi Provincial Hospital and, by extension, other Zimbabwean public hospitals.

1.2 Problem Statement

1.2.1 Ideal Situation

In an ideal healthcare environment, every patient who enters a public hospital should be swiftly registered, electronically triaged according to the severity of their condition, and directed to the appropriate service area with minimal waiting time. Real-time digital dashboards should enable clinical staff and hospital administrators to monitor patient flow across all departments, identify bottlenecks instantly, and dynamically reallocate resources to prevent congestion. Patients should be notified when their turn is approaching via WhatsApp or web notification, even if they have temporarily left the hospital premises. Critically ill patients should be automatically flagged and fast-tracked, and all patient journey data should be recorded digitally for reporting, quality improvement, and health system planning.

1.2.2 Reality

The current reality at Chinhoyi Provincial Hospital, as evidenced by the case study, is significantly different. Patient registration, triage classification, and queue management are entirely manual and paper-based. There is no system to digitally track a patient's journey from the point of entry through triage, consultation, investigations, pharmacy, and discharge. Triage officers rely on visual assessment and handwritten records to assign patients to the colour-coded priority categories (Red, Yellow, Green, Black), with no digital support to flag deteriorating patients or compute waiting times.

The OPD flow map, while conceptually well-designed, is not supported by any digital infrastructure. Queue bottlenecks, such as delays at the laboratory (20 minutes) or radiology (30 minutes), cannot be monitored in real time. The OI Clinic manages multiple complex patient streams through paper-based files with no automated scheduling or queue prioritisation. Patient sources include both unreferred walk-in cases and referred cases from clinics, peripheral hospitals, and private practitioners, adding further complexity to patient flow management. Critically, patients who leave the hospital premises while waiting have no way of being notified when it is their turn, leading to no-shows and further disruption of the queue.

1.2.3 Consequences

The consequences of the current manual system include:

- Prolonged patient waiting times that regularly exceed 3-5 hours for a standard OPD visit, far beyond the expected 80 minutes under ideal conditions.
- Unmanaged congestion in OPD and casualty areas, with risk of adverse clinical outcomes when critically ill patients are not prioritised.
- Staff burnout due to manual administrative burden, including physical movement between departments to coordinate patient referrals.
- Inability to generate real-time or historical performance reports, preventing evidence-based staffing and resource decisions.
- Loss of institutional data due to paper records being misfiled or damaged.
- No-shows caused by patients who leave the premises without knowing when their turn will come, as there is no notification mechanism.
- Patients from referred facilities who arrive without prior digital notification experience additional, avoidable delays.

1.2.4 Proposed Solution

This project proposes the development of WaitLess, a web-based digital queue management system tailored for Zimbabwean public hospitals. The system will digitise patient registration and triage, automate patient-to-department routing based on triage priority, provide real-time queue dashboards for clinical and administrative staff, generate waiting time estimates, and notify patients when their turn is approaching via WhatsApp or web push notification. It will also produce automated reports for hospital management. The system will be designed to function on low-bandwidth networks using locally hosted infrastructure, making it suitable for resource-constrained hospital settings.

Problem Statement: Chinhoyi Provincial Hospital, like most public hospitals in Zimbabwe, manages patient registration, triage, and queue flow using entirely manual, paper-based processes. No digital system exists to track patient movement from entry through triage, consultation, investigations, and discharge, nor to monitor waiting times, alert staff to congestion, or notify patients of their queue status. This results in inequitable service delivery, preventable delays in care, and loss of operational data. A web-based digital queue management system tailored to Zimbabwe's triage protocols and hospital workflows would enable real-time patient tracking, automated priority routing, patient notifications via WhatsApp and web push, and data-driven hospital management.

1.3 Project Aims and Objectives

1.3.1 Aim

The aim of this project is to design, develop, and evaluate a web-based digital queue management system for Zimbabwean public hospitals, using Chinhoyi Provincial Hospital as the primary case study.

1.3.2 Objectives

The following objectives have been defined to guide the achievement of the project aim:

1. To analyse current patient flow, triage, and queue management processes at Chinhoyi Provincial Hospital and identify key system requirements, including the need for a patient notification mechanism, by May 2026.
2. To design and develop a web-based digital queue management system that supports colour-coded triage prioritisation, multi-department patient routing, real-time queue monitoring, and a WhatsApp and web push patient notification engine, by August 2026.
3. To evaluate the system's usability, efficiency, and suitability for deployment in low-bandwidth, resource-limited Zimbabwean hospital settings, including patient satisfaction with the notification system, by October 2026.

1.4 Significance of the Study

This study is significant in terms of societal impact, healthcare service improvement, academic contribution, and alignment with national digital health objectives.

- **Societal Impact:** Long patient waiting times in Zimbabwean public hospitals remain a major healthcare challenge. By digitising and streamlining queue management processes at Chinhoyi Provincial Hospital, the proposed system has the potential to improve patient experience and reduce congestion in waiting areas. The integration of patient notifications further allows patients to temporarily leave waiting areas without losing their queue position.
- **Healthcare Worker Efficiency:** The automation of queue coordination tasks reduces the administrative burden currently placed on nurses and hospital staff. This allows healthcare personnel to focus more on clinical responsibilities, thereby improving operational efficiency and potentially enhancing job satisfaction.
- **Evidence-Based Management:** The system generates both real-time and historical data relating to patient flow, waiting times, and departmental throughput. Such information can support hospital management in making informed decisions regarding staffing, scheduling, and resource allocation.
- **Reduction of Missed Queue Turns:** The integration of WhatsApp and web push notifications addresses the common problem of patients missing their queue turn after temporarily leaving the hospital premises. By notifying patients when their turn is approaching, the system improves queue continuity and reduces service disruptions.
- **Academic Contribution:** The study contributes to the growing body of knowledge in digital healthcare systems within sub-Saharan Africa, particularly in Zimbabwe. The project may serve as a reference model for similar queue management implementations in other public healthcare institutions.

- **Alignment with National Health Objectives:** The project aligns with Zimbabwe's National Health Strategy 2021–2025 and the Digital Economy Blueprint, both of which encourage the adoption of digital technologies to improve healthcare service delivery and patient experience in public institutions.

1.5 Limitations and Delimitations of the Study

1.5.1 Limitations of the Study

The following limitations are acknowledged in this study:

- The case study was conducted at a single hospital, Chinhoyi Provincial Hospital, which may not fully represent the diversity of patient flow processes and infrastructure conditions across all Zimbabwean public hospitals.
- Intermittent internet connectivity at Chinhoyi Provincial Hospital may affect the WhatsApp notification and web push features of the system. However, the core queue management system is designed to function offline within the hospital local area network, with notifications queued for delivery once internet connectivity is restored.
- Variations in patient and staff digital literacy levels may affect system usability, thereby necessitating simple user interfaces and staff orientation sessions.
- Access to hospital records and patient flow data for baseline comparison may be limited due to the manual nature of existing systems and data privacy considerations.
- Intermittent power supply and limited hardware resources at the hospital may affect system deployment and testing activities.
- The limited timeframe allocated for the final-year project may restrict the depth of user acceptance testing and system evaluation conducted.

1.5.2 Delimitations of the Study

The study is delimited as follows:

- The system focuses exclusively on Chinhoyi Provincial Hospital as the case study institution. Generalisation to other hospitals is proposed but falls outside the scope of this project.
- The system covers five departments: Outpatient Department (OPD), Casualty/Triage, Pharmacy, Laboratory/Radiology, and Opportunistic Infection (OI) Clinic. Inpatient ward management and theatre scheduling are excluded from the scope of the project.
- Patient notifications are limited to WhatsApp Business API notifications and web push notifications only. SMS notifications and dedicated mobile applications are not included in this phase of the project.
- Queue display monitors for waiting areas are identified as possible future enhancements but are not part of the current project deliverables.
- The system does not integrate with existing Hospital Information Systems (HIS) or Electronic Health Records (EHR) platforms in this phase.
- The project focuses on prototype development and pilot evaluation rather than full-scale production deployment.

1.6 Definition of terms

Queue Management System:
A digital system used to organise, monitor, and manage the flow of patients waiting for healthcare services in order to improve efficiency and reduce waiting times.

Triage:
The process of prioritising patients according to the severity and urgency of their medical condition to ensure timely medical attention.

Outpatient Department (OPD):
A hospital department that provides registration, consultation, and treatment services to patients who do not require overnight admission.

Casualty Department:
The hospital department responsible for the initial reception, triage, and emergency treatment of patients requiring urgent medical attention.

Opportunistic Infection (OI) Clinic:
A specialised hospital clinic that provides care for HIV/AIDS patients, including antiretroviral therapy resupply, TB services, viral load testing, cervical cancer screening, and counselling.

Digital Health System:
A technology-based healthcare solution designed to improve healthcare service delivery, patient management, and operational efficiency through the use of digital tools and platforms.

Web Push Notification:
A browser-based notification sent from a web application to a user's device to provide alerts or updates, even when the application is not actively open.

WhatsApp Business API:
A communication platform provided by Meta that enables businesses and organisations to send automated messages and notifications to users through WhatsApp.

Local Area Network (LAN):
A computer network that connects devices within a limited geographical area such as a hospital, office, or campus to enable communication and resource sharing.

Patient Flow:
The movement of patients through different stages of healthcare service delivery, from registration and triage to consultation, treatment, and discharge.

Real-Time System:
A system capable of processing and updating information immediately or with minimal delay as events occur.

Hospital Information System (HIS):
An integrated digital system used by healthcare institutions to manage administrative, financial, and clinical operations.

Electronic Health Record (EHR):
A digital version of a patient's medical history and health information maintained over time by healthcare providers.

WebSocket:
A communication protocol that enables continuous two-way communication between a client application and a server in real time.

Prototype:
An initial working version of a system developed for testing, evaluation, and demonstration purposes before full deployment.

Application Programming Interface (API):
A set of rules and protocols that allows different software applications or system components to communicate and exchange data with each other.

Departmental Queue:
A queue assigned to a specific hospital department where patients wait to receive services related to that department.

Patient Notification System:
A subsystem responsible for sending alerts and updates to patients regarding their queue status, appointment status, or service availability.

1.7 Conclusion

This chapter introduced the development of WaitLess, a digital queue management system designed for Zimbabwean public hospitals, with Chinhoyi Provincial Hospital serving as the case study institution. The chapter discussed the existing manual patient flow and triage processes currently used at the hospital and identified key inefficiencies associated with the absence of a digital queue management system.

The chapter further highlighted the significance of integrating digital queue coordination and patient notification mechanisms into public healthcare service delivery. Particular emphasis was placed on the use of WhatsApp Business API notifications and web push notifications to improve patient convenience and reduce missed queue turns without introducing recurring messaging costs to the hospital.

The aim, objectives, significance, limitations, and delimitations of the study were also presented to establish the boundaries and direction of the project. Collectively, these elements provide the foundation for the design and development of an affordable, scalable, and contextually appropriate queue management solution for Zimbabwean public healthcare institutions.

The next chapter reviews related literature and existing digital healthcare solutions relevant to queue management systems, patient flow optimisation, and healthcare digitalisation.

CHAPTER 2: LITERATURE REVIEW

2.0 Introduction

This chapter presents a review of literature relevant to the development of WaitLess. It examines existing scholarship on queue management theory in healthcare, digital queue management system implementations in hospital settings, patient notification mechanisms, digital health adoption in Zimbabwe and the wider Sub-Saharan African region, design science research as a methodological foundation for information systems projects, and the technology frameworks suited to real-time web applications. The chapter closes by synthesising the gaps that emerge from this body of work and explaining how WaitLess is positioned to address them.

The chapter is organised thematically. Section 2.1 begins with the theoretical foundations of queue management. Section 2.2 reviews specific digital queue management system implementations in hospital contexts, both globally and within Sub-Saharan Africa. Section 2.3 focuses on patient notification systems, a central feature of WaitLess. Section 2.4 examines digital health adoption in Zimbabwe. Section 2.5 discusses design science research as the methodological foundation adopted for this project, and Section 2.6 reviews the technology frameworks relevant to its implementation. Section 2.7 reviews existing commercial queue management systems, and Section 2.8 synthesises the gaps identified across the reviewed literature. Section 2.9 concludes the chapter.

2.1 Queue Management Theory in Healthcare

Queue management, as a field of academic enquiry, draws on queuing theory, a branch of operations research concerned with the mathematical study of waiting lines. Preater (2002) traced the growing application of queuing theory within healthcare and medicine, arguing that queue modelling provides a foundational analytical tool for understanding and managing patient flow within hospitals. This early framing established queue management not merely as an operational inconvenience to be minimised, but as a legitimate object of systematic study with direct implications for clinical safety and service quality.

Building on this foundation, Cayirli and Veral (2003) conducted a comprehensive review of outpatient scheduling and appointment systems literature. Their review concluded that the primary drivers of patient waiting time were unpredictable patient arrival patterns, variable service durations, and the absence of real-time monitoring tools, and argued that digital scheduling and queue management systems were essential tools for addressing these inefficiencies.

Fomundam and Herrmann (2007) extended this line of enquiry through a broader survey of queuing theory applications across the healthcare sector. Their survey concluded that the primary barriers to effective queue management in public hospitals were the lack of real-time data visibility and the absence of coordinated patient tracking systems spanning multiple departments. This finding mirrors the conditions documented during the case study conducted at Chinhoyi Provincial Hospital, where patient movement between the OPD, laboratory, radiology, and pharmacy is coordinated entirely through handwritten records with no shared visibility across departments.

Taken together, these three works establish queue management as a well-theorised problem domain, but one whose proposed solutions have historically remained largely conceptual or confined to high-income, well-resourced healthcare settings. None of the three studies

reviewed in this section report on an implemented system evaluated in a low-resource, Sub-Saharan African hospital, a gap addressed by the empirical implementations reviewed in the following section.

2.2 Digital Queue Management Systems in Hospitals

Where Section 2.1 established the theoretical case for digital queue management, a growing body of empirical literature has evaluated implemented systems directly. Egbunu et al. (2020) developed a web-based centralised queue control system for non-tertiary hospitals in Nigeria, a healthcare and infrastructural context closely comparable to Zimbabwe. Their system was built using ASP.NET and Microsoft SQL Server and was evaluated using operational data drawn from hospitals in Benue State, Nigeria. The evaluation reported a reduced number of patients present in the hospital at any given time, reduced physical crowding, and improved overall service delivery. However, the Egbunu et al. system did not incorporate a formal triage prioritisation module aligned with colour-coded severity classification, nor did it include a patient notification component; patients were still required to remain physically present within the facility to retain their place in the queue. Both of these are core features of WaitLess, and their absence from the Egbunu et al. system represents a direct gap that this project addresses.

Bidari et al. (2021) provided some of the strongest causal evidence available in this literature through a prospective randomised controlled trial evaluating the effect of a queue management system on patient satisfaction in a hospital emergency department. The trial involved 236 patients allocated to intervention and control groups. The results showed that both actual and perceived waiting times were significantly lower in the intervention group, and that patient satisfaction improved markedly with the introduction of the queue management system. The randomised design of this study lends particular methodological weight to the broader claim, made throughout this literature, that queue management interventions produce measurable improvements in patient experience rather than merely a perception of improvement.

Wijeratne et al. (2019) evaluated a token-based electronic queue system implemented at a government hospital in Sri Lanka, reporting a 35% reduction in average patient waiting time alongside a significant improvement in patient satisfaction scores. Notably, the system evaluated by Wijeratne et al. incorporated automated patient notifications as a core design component rather than an optional add-on, offering evidence that patient-facing alerts are integral to the effectiveness of a queue management system rather than a peripheral convenience feature. This finding directly supports the architectural decision, central to WaitLess, to treat the notification engine as a first-class system component.

Within the African context specifically, an evaluation of technical and workflow interventions at Kilimanjaro Christian Medical Centre (KCMC) in Tanzania, reported by Mboya et al. (2024), found that KCMC had historically recorded the longest outpatient waiting times among Tanzania's zonal and national referral hospitals, with patients waiting an average of six hours to receive OPD services. In response, the hospital implemented a multi-year programme of interventions, including the expansion of physical infrastructure, the streamlining of patient flow processes, the deployment of patient ushers, and increases in clinical staffing, which together produced a substantial and sustained reduction in patient waiting times. This case is instructive for the present study in two respects: first, it confirms that resource-constrained East African referral hospitals can achieve measurable waiting-time reductions through deliberate intervention; second, it demonstrates that the most effective intervention documented in a comparable regional setting relied primarily on process redesign and staffing changes rather than on a dedicated, purpose-built digital queue management platform of the kind proposed in this project.

Collectively, these four studies demonstrate that digital and process-based queue management interventions are both feasible and effective across a range of healthcare settings, from high-income emergency departments to resource-constrained African referral hospitals. However, only Wijeratne et al. (2019) explicitly integrated patient notifications, and none of the four studies combined patient notification with a colour-coded triage prioritisation model of the kind required by Zimbabwean public hospitals. This combination represents a specific, unaddressed gap in the reviewed literature.

2.3 Patient Notification Systems in Healthcare

A recurring theme across the queue management literature reviewed in Section 2.2 is the growing recognition that patient notification is not a peripheral feature but a core determinant of system effectiveness. Despite this, the specific mechanism of notification varies considerably across implementations, and few studies address notification design in a manner appropriate to a resource-constrained, low-connectivity environment such as Chinhoyi Provincial Hospital.

The case study conducted at Chinhoyi Provincial Hospital identified the complete absence of any mechanism to notify patients of their queue status, particularly patients who leave hospital premises while waiting, as a critical gap in the existing manual process. WaitLess addresses this gap through two complementary notification channels selected specifically to minimise or eliminate ongoing cost to the hospital.

The primary channel is the WhatsApp Business API. According to Meta (2024), the WhatsApp Business Platform's service conversation pricing model allows a business or organisation to reply to a patient-initiated message free of charge for a 24-hour window following that message. Applied to a hospital queue context, this means that if a patient sends an initiating message to the hospital's WhatsApp Business number at the point of registration, all subsequent queue status notifications sent to that patient during a single-day visit fall within the free service window. Since a typical OPD visit at Chinhoyi Provincial Hospital is completed within a single day, this pricing structure allows WaitLess to deliver WhatsApp notifications at no recurring cost to the hospital, directly addressing one of the key adoption barriers identified by Gombe et al. (2019) in relation to digital health tools in Zimbabwe, namely constrained hospital IT budgets.

The secondary channel, web push notification, operates entirely within the system's own infrastructure and therefore incurs no third-party messaging costs at all. This channel is particularly suited to patients who register in advance through the web application, or who access the system from a personal device while on the hospital premises.

A further design consideration, raised but placed outside the current project scope, is the use of physical queue display monitors positioned at departmental waiting areas. Such displays are already common in Zimbabwean retail and banking environments and would extend queue visibility to patients without phone access. Because this feature requires only a browser-capable display device rather than new backend infrastructure, it is identified as a natural next-phase enhancement following a successful pilot of the notification engine, rather than as a limitation of the underlying system design.

The design of the WaitLess notification engine therefore responds directly to two distinct gaps identified elsewhere in this chapter: the general under-prioritisation of patient notification observed in Section 2.2, and the specific absence of any notification mechanism documented in the Chinhoyi Provincial Hospital case study.

2.4 Digital Health Adoption in Zimbabwe and Sub-Saharan Africa

Despite the demonstrated benefits of digital queue management systems reviewed in Section 2.2, adoption of digital health tools more broadly across Sub-Saharan Africa, and within Zimbabwe specifically, remains uneven. Gombe et al. (2019) conducted a mixed-methods study analysing the barriers to health information system adoption in Zimbabwe. Their study identified three primary obstacles: inadequate infrastructure, limited information technology skills among healthcare workers, and a lack of locally developed, context-appropriate solutions.

This last barrier is of particular relevance to the present study. Much of the digital queue management literature reviewed in Section 2.2, and the commercial systems reviewed in Section 2.7, originates from or is designed for higher-resource contexts. Systems built for reliable broadband connectivity, well-staffed IT departments, or predictable electricity supply are not necessarily transferable, without substantial redesign, to a Zimbabwean provincial hospital operating under intermittent connectivity and constrained infrastructure. Gombe et al.'s finding therefore provides direct empirical support for the project's core methodological decision to base system requirements on a physical case study conducted at Chinhoyi Provincial Hospital, rather than adapting an existing solution designed elsewhere.

The infrastructural barriers identified by Gombe et al. also align closely with the operating conditions documented at Chinhoyi Provincial Hospital, where intermittent internet connectivity and constrained hardware resources were identified as limitations that WaitLess must accommodate by design, rather than as edge cases to be handled separately.

2.5 Design Science Research as a Methodological Foundation

Beyond the substantive literature on queue management and digital health, this project also draws on methodological literature to justify its choice of research approach, which is elaborated fully in Chapter 3. Hevner et al. (2004) established a foundational framework for design science research (DSR) within the information systems discipline. They argued that DSR involves the creation of purposeful artefacts, such as software systems, models, or methods, that address specific, real-world organisational problems, and that the utility, quality, and efficacy of such artefacts must be rigorously and iteratively evaluated rather than simply asserted.

Bisandu (2019) further contextualised design science research specifically within computer science and information systems research, emphasising its particular relevance for projects that both develop and evaluate a software artefact as a direct response to a documented organisational problem. This framing maps closely onto the structure of the present project: the physical case study conducted at Chinhoyi Provincial Hospital documents a specific organisational problem, and the WaitLess system constitutes the designed artefact proposed as its solution.

The relevance of DSR to this project lies not only in its theoretical fit but in the explicit emphasis both Hevner et al. (2004) and Bisandu (2019) place on iterative evaluation. This has directly informed the decision to structure system development around distinct, evaluable phases, namely requirements analysis, design, implementation, testing, and evaluation, as detailed in Chapter 3, rather than pursuing a single-pass build with no structured evaluation checkpoint.

2.6 Technology Frameworks for Real-Time Web Applications

The implementation of WaitLess depends on technology choices capable of supporting real-time, concurrent updates across multiple hospital departments simultaneously. React and Node.js have emerged as a widely adopted full-stack combination for building this class of application. Aggarwal (2018) demonstrated that React's component-based architecture is particularly well suited to building live dashboard interfaces that must reflect dynamic, frequently changing data, a requirement central to the WaitLess staff dashboard, which must reflect queue state changes across the OPD, Casualty, Pharmacy, Laboratory/Radiology, and OI Clinic departments in near real time.

On the server side, Tilkov and Vinoski (2010) argued that Node.js's event-driven, non-blocking input/output model makes it well suited to handling large numbers of concurrent, lightweight updates of the kind generated by multiple hospital departments simultaneously reporting queue state changes. This architectural characteristic directly informs the choice of Node.js with Express.js and WebSocket support as the backend foundation for WaitLess, since a traditional synchronous, thread-per-request server architecture would be comparatively poorly suited to sustaining many simultaneous, low-latency department-to-dashboard update streams on the modest server hardware available within a hospital local area network.

While Aggarwal (2018) and Tilkov and Vinoski (2010) establish the general suitability of this stack for real-time applications, neither study addresses deployment within a low-bandwidth, intermittently connected environment of the kind found at Chinhoyi Provincial Hospital. This is not a shortcoming of either work, both of which concern general web application architecture rather than healthcare deployment specifically, but it does mean that the low-bandwidth, offline-tolerant configuration of WaitLess's technology stack, detailed in Chapter 3, extends beyond what either source directly evaluates.

2.7 Review of Commercial Queue Management Systems

Beyond the academic literature reviewed above, a review of existing commercial queue management systems provides additional context for the gap this project addresses. Established commercial platforms such as Qmatic, Qtrac, and Wavetec are feature-rich, mature products with a strong track record of deployment in high-income country settings, including retail, banking, and some private healthcare environments. However, three characteristics limit their suitability for a Zimbabwean public hospital context. First, licensing and deployment costs are typically structured for well-resourced private or high-income public institutions, placing them outside the budget of a Zimbabwean public hospital such as Chinhoyi Provincial Hospital. Second, these platforms generally assume consistent, high-bandwidth internet connectivity, an assumption that does not hold reliably at Chinhoyi Provincial Hospital. Third, and most significantly for this project, none of these commercial platforms are designed around the specific colour-coded triage protocols (Red, Yellow, Green, Black) or the multi-department referral workflows that characterise Zimbabwean public hospital operations; their prioritisation and routing logic is generic rather than aligned to this clinical context.

This review of commercial systems is necessarily limited to publicly available product information rather than independent, peer-reviewed evaluation, and should be read as market context rather than as literature in the academic sense. Even accounting for this limitation, the pattern is consistent with the academic literature reviewed in Sections 2.1 and 2.2: mature, well-resourced solutions exist, but none combine affordability, offline tolerance, and Zimbabwean-triage-aware routing in a single system.

2.8 Identified Gaps in the Literature

The literature reviewed in this chapter establishes that queue management is a well-theorised problem (Section 2.1), that digital and process-based interventions have been implemented and evaluated with measurable success across a range of healthcare settings (Section 2.2), that patient notification is increasingly recognised as a core rather than peripheral system feature (Section 2.3), and that infrastructural and contextual barriers continue to limit digital health adoption specifically within Zimbabwe (Section 2.4). Read together, this body of work reveals a specific combination of gaps that no single reviewed source addresses simultaneously.

First, none of the digital queue management implementations reviewed in Section 2.2, including Egbunu et al.'s (2020) Nigerian system and the KCMC intervention reported by Mboya et al. (2024), combine automated patient notification with colour-coded triage prioritisation aligned to a four-tier severity classification of the kind used at Chinhoyi Provincial Hospital. Wijeratne et al. (2019) incorporated notifications but not triage-aware routing; Egbunu et al. (2020) addressed multi-department routing but not notifications.

Second, no reviewed implementation was explicitly designed for zero-recurring-cost patient notification in a low-resource public hospital setting. The token-based and application-based notification approaches described in the literature generally assume either dedicated hardware or standard messaging costs; none leverage a patient-initiated messaging-platform pricing structure, of the kind Meta's (2024) WhatsApp Business service conversation policy makes possible, to eliminate recurring notification costs.

Third, the commercial systems reviewed in Section 2.7 demonstrate that feature-complete queue management platforms exist, but at a cost and connectivity requirement that places them outside the reach of Zimbabwean public hospitals, while the academic literature on affordable, locally developed digital health tools (Section 2.4) has not, to date, produced a queue management system specifically aligned to Zimbabwean triage protocols.

Fourth, and more generally, the reviewed African implementations (Egbunu et al., 2020; Mboya et al., 2024) demonstrate that digital and process interventions can succeed in resource-constrained regional hospital settings, but neither combines this feasibility with the specific notification architecture WaitLess proposes.

WaitLess is positioned directly within this combined gap: a web-based digital queue management system that integrates colour-coded triage-aware routing, real-time multi-department queue visibility, and a zero-recurring-cost patient notification engine, purpose-built for the infrastructural and budgetary realities of a Zimbabwean public hospital.

2.9 Conclusion

This chapter has reviewed literature across six interconnected areas: the theoretical foundations of queue management in healthcare, empirical implementations of digital queue management systems in hospital settings, patient notification mechanisms, digital health adoption within Zimbabwe and the wider Sub-Saharan African region, design science research as a methodological foundation, and the technology frameworks suited to real-time web application development. A review of existing commercial queue management platforms provided additional market context.

Across this literature, a consistent pattern emerged: digital interventions in queue management are both feasible and effective, but no single reviewed system combines triage-aware patient routing, real-time multi-department visibility, and a zero-recurring-cost patient notification engine within a system designed for the infrastructural and budgetary constraints of a Zimbabwean public hospital. This combination of gaps, detailed in Section 2.8, defines the specific contribution WaitLess is intended to make.

The following chapter presents the methodology adopted to design, develop, and evaluate WaitLess, structured according to the design science research framework established by Hevner et al. (2004) and discussed in Section 2.5 of this chapter.